

Process Management

Program

- Set of instructions given to the computer --> Executable file.
- Program --> Sectioned binary --> "objdump" & "readelf".
 - Exe header --> Magic number, Address of entry-point function, Information about all sections. (objdump -h program.out)
 - Text --> Machine level code (objdump -S program.out)
 - Data --> Global and Static variables (Initialized)
 - BSS --> Global and Static variables (Uninitialized)
 - RoData --> String constants
 - Symbol Table --> Information about the symbols (Name, Size, section, Flags, Address) (objdump -t program.out)
- Program (Executable File) Format
 - Windows -- PE
 - Linux -- ELF
- Program are stored on disk (storage).

Process

- Program under execution
- Process execute in RAM.
- Process control block contains information about the process (required for the execution of process).
 - Process id
 - Exit status
 - 0 - Indicate successful execution
 - Non-zero - Indicate failure
 - Scheduling information (State, Priority, Sched algorithm, Time, ...)
 - Memory information (Base & Limit, Segment table, or Page table)
 - File information (Open files, Current directory, ...)
 - IPC information (Signals, ...)
 - Execution context (Values of CPU registers)
 - Kernel stack
- PCB is also called as process descriptor (PD), uarea (UNIX), or task_struct (Linux).
- In Linux size of task_struct is approx 4KB

Process

- Process is program in execution.
- Process has multiple sections i.e. text, data, rodata, heap, stack. ... into user space and its metadata is stored into kernel space in form of PCB struct.
- PCB contains
 - id, exit status,
 - scheduling info (state, priority, time left, scheduling policy, ...),
 - files info (current directory, root directory, open file descriptor table, ...),
 - memory information (base & limit, segment table, or page table),

- ipc information (signals, ...),
- execution context, kernel stack, ...

Disk/partition naming conventions

- * Windows:
 - * Disks are named as disk0, disk1, ...
 - * partitions are named as drives i.e. C:, D:, E:, ...
- * Linux:
 - * Disks are named as /dev/sda, /dev/sdb, /dev/sdc, etc.
 - * Partitions per disk are named as
 - * sda partitions: sda1, sda2, sda3, ...
 - * sdb partitions: sdb1, ...

Types of files

- Regular file (-)
- Directory file (d)
- Link file (l)
- Socket file (s)
- Pipe file (p)
- Character Special file (c)
- Block Special file (b)

Process Life Cycle

Process States

- New
 - New process PCB is created and added into job queue. PCB is initialized and process get ready for execution.
- Ready
 - The ready process is added into the ready queue. Scheduler pick a process for scheduling from ready queue and dispatch it on CPU.
- Running
 - The process runs on CPU. If process keeps running on CPU, the timer interrupt is used to forcibly put it into ready state and allocate CPU time to other process.
- Waiting
 - If running process request for IO device, the process waits for completion of the IO. The waiting state is also called as sleeping or blocked state.
- Terminated
 - If running process exits, it is terminated.

Redirection

- By default every process opens 3 file descriptors

- fd = 0 -> standard input to a program
- fd = 1 -> standard output to a program
- fd = 2 -> standard error to a program
- You can redirect each of these independently.
- According to direction of redirection, there are three types
- Input redirection
 - "<" is used for input redirection
- Output redirection
 - ">" is used for output redirection
- Error redirection
 - "2>" is used for error redirection

Pipe

- Using pipe, we can redirect output of any command to the input of any other command.
- Two processes are connected using pipe operator (|).
- Two processes runs simultaneously and are automatically rescheduled as data flows between them.
- If you don't use pipes, you must use several steps to do single task.
- E.g.
 - who | wc